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Professional Acoustic Services

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Acoustic Design Report

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Extension and Remodelling:

Penkford School

Red Bank Community House

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1.0 Introduction

Azymuth Acoustics UK is appointed by Galliford Try to provide design advice for the proposed extension and remodelling of Penkford School at Red Bank Community House. The scheme generally comprises creation of key skills classrooms, a 2000m³ multi-purpose hall, a new kitchen and dining area as well as staff and ancillary rooms.

Remodelling includes the construction of new partition walls to form new rooms across the existing building. The following schedules have been developed with reference to Acoustic Design of Schools: Performance Standards 2015 (Building Bulletin 93) and other relevant guidance.

The school is located off Newton Road A49, a busy road linking the M62 and Newton-le-Willows.

This acoustic design report includes sections covering the following aspects:

- Results of an initial baseline noise survey undertaken in the vicinity of the proposed extension.
- Detailing of acoustic assessment criteria relevant to the scheme.
- Recommendations on sound insulation of the building envelope and internal elements between newly formed spaces.
- Comments on plant and ventilation in newly formed spaces.
- Recommendations for reverberation control in teaching and learning spaces in order to facilitate a comfortable acoustic environment.

2.0 Baseline Noise Survey

Azymuth Acoustics undertook a baseline noise survey of the site with the aim of providing specifications of building envelope elements suitable to mitigate noise break-in.

2.1 Measurement Procedures

The environmental noise survey was carried out in compliance with methodology described in BS 7445. The instrumentation used for the environmental noise survey consisted of a SvanTek 959 type 1 precision sound level meter.

The equipment was calibrated before and after the noise measurements. The sound level meter recorded A-weighted (fast response) noise levels as well as octave bands noise levels for all measurements.

The survey was undertaken between 11:18 to 12:45 hrs on Wednesday 8th January 2020. The survey covered a representative period of school opening hours.

Ambient noise measurements were undertaken using the sound level meter microphone at the locations shown in Figure 1 below.

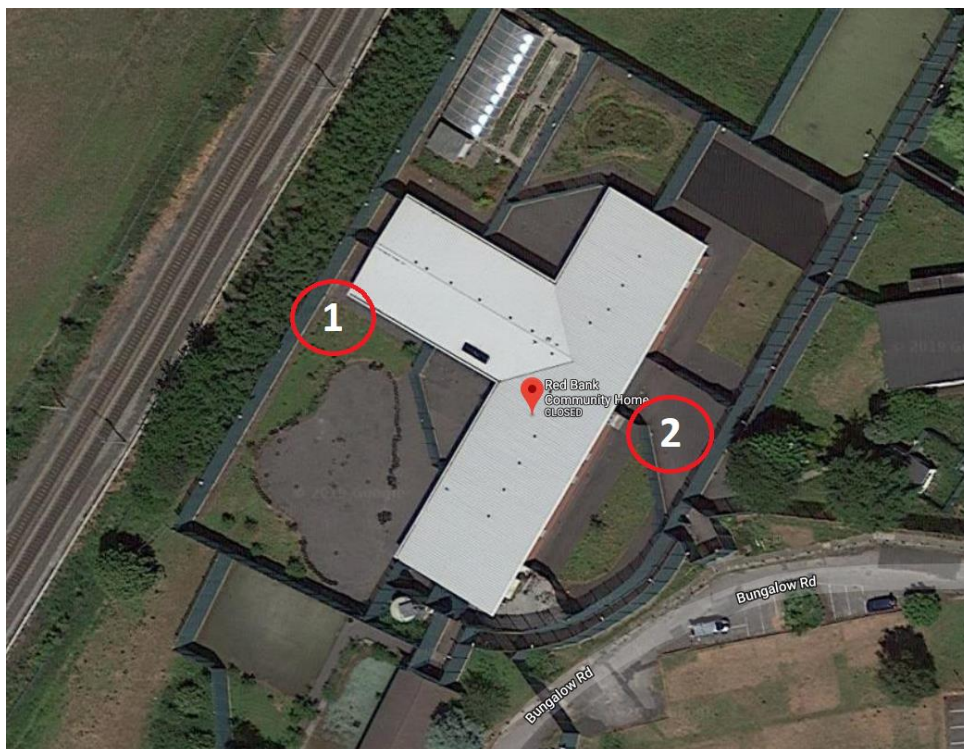


Figure 1: Survey measurement locations shown numbered on aerial photograph

2.2 Results of Noise Measurements

Full results of the noise levels recorded during the course of the survey are included in Appendix B of this report. The following table summarises the results of the noise measurements undertaken at the site in terms of the average daytime (09:00 to 15:00 hours) statistical noise levels.

Position	Time	L _{Amax}	L _{Aeq}	L _{A1}	L _{A10}	L _{A50}	L _{A90}
1) Rear of existing building (15m from trainline)	11:18 – 12:23	63.0	56.7	59.7	47.9	42.9	41.5
2) Front of existing building.	12:25 – 12:45	59.4	46.3	55.3	46.1	43.4	41.9

Table 1: Results of Azymuth Acoustics UK noise measurements, dB

2.3 Description of Noise Climate

During the daytime the principal source of noise in the vicinity of the site is rail traffic on the West Coast Mainline railway. This section of the West Coast Mainline is used predominantly by a mixture of intercity and freight trains. The intercity trains are themselves a mixture of Virgin Pendolino trains (serving the London Euston to Glasgow route) and Virgin Voyager trains (serving the Birmingham to Glasgow and Edinburgh routes).

The proposed school is located near to Winwick Road (the A49). This is a relatively busy road which links Warrington / Winwick with the A580 (East Lancashire) trunk road. Noise from vehicles using Winwick Road together with farm vehicles / machinery on the surrounding fields contributes to the general background noise level in the area.

3.0 Assessment Criteria

In order to assess the extent of any measures required in order to comply with suitable conditions relating to potential noise sources, Azymuth Acoustics has reviewed various guidance documents and standards, these include:

- BB93 Acoustic Design of Schools: Performance Standards 2015
- British Standard 4142: 2014

3.1 BB93 Acoustic Design of Schools: Performance Standards 2015

Building Bulletin 93 2015 replaces the Acoustic Performance Standards document of 2013 and sets performance standards for the acoustics of new educational buildings.

The document provides detailed guidance on the following aspects:

- Indoor ambient noise levels in unoccupied spaces, taking into consideration both environmental noise break-in and ventilation / plant noise within each space.
- Airborne and impact sound insulation between teaching spaces, ancillary rooms and to circulation areas etc.
- Control of reverberation in teaching, study, sports and circulation spaces

3.2 British Standard 4142: 2014

The test that is generally applied to assess the potential for noise from industrial and commercial installations to give rise to community response is contained in British Standard BS 4142: 2014. This standard predicts the likelihood of complaint due to noise from a new or modified source.

BS 4142 describes a method for determining the specific source and background noise levels outside residential buildings and for assessing whether the noise is likely to give rise to complaints from the occupants. The specific noise level from the commercial source is rated based on any tonal or intermittent characteristics of the sound. BS 4142 assessment methodology involves comparing the existing background level with the predicted noise from the new development. This predicted level is adjusted to allow for any tonal or impulsive characteristics and is called the Rating Level. The difference between the two levels can be used to indicate the likelihood of complaints arising, e.g. a predicted level of 10dB greater than the background level indicates a significant adverse impact; an increase of around 5dB indicates an adverse impact and where the rating level does not exceed the background sound level (BSL) this is an indication of the source having a low impact.

4.0 External Envelope

4.1 Target Indoor Ambient Noise Levels

The following table provides a summary of internal noise limits for various rooms applicable to the scheme.

Room Type (Refurbishment)	Noise limit (break-in), unoccupied spaces (BB93) $L_{Aeq,30min}$
Assembly hall, multi-purpose hall (drama, PE, audio/visual presentations, assembly, occasional music)	40dBA
<i>Secondary school:</i> classroom, general teaching area, seminar room, tutorial room, language laboratory	40dBA
Office, medical room, staff room	45dBA
Foyer, circulation spaces	45dBA

Table 2: Targets for indoor ambient noise levels

4.2 External Walls

The minimum recommended sound insulation for the external walls and/or cladding is R_w 49dB. It is understood the existing external walls across the development are constructed using outer leaf facing brickwork with 140mm blockwork inner leaf, 100mm cavity partly filled with 60mm insulation. The proposed specification will provide a sound insulation rating significantly greater than R_w 49dB and as such the external wall construction details would be deemed to be satisfactory.

4.3 External Glazing and Doors

Table 4 below details the required external glazing specifications, based on the proposed layout and results of the baseline noise survey.

Area	Required reduction from glazing	Recommended specification
All new external windows to all areas / façades	Min. R_w 31dB and R_{w+Ctr} 27dB	Good quality thermal double glazing
All new external doors	Min. R_w 31dB	Good quality double glazed door with suitable airtight threshold seal

Table 3: Recommended external glazing specifications

Incorporation of the recommendations above would result in compliance (windows closed) with indoor ambient noise level limits described in Building Bulletin 93.

4.4 Roof Construction

The proposed roof is understood to be a Kingspan Kingzip IP (KS500 / 1000) standing seam construction with mineral wool insulation core on profiled metal deck.

Classrooms and other pupil areas would have plasterboard ceilings at high level, with circulation areas incorporating a mineral tile grid system.

The overall proposed construction would provide sufficient airborne sound insulation, based on measured external noise levels in the vicinity. It would be expected that rain hammering noise would be sufficiently mitigated based on a mineral wool insulated roof panel and internal ceiling below.

Where no ceiling is proposed to teaching spaces (e.g. multi-purpose hall) incorporation of SRS Raincheck (with 40% roof coverage) on top of the Kingspan system should be considered, which would mitigate potentially higher internal noise levels during periods of heavy rain.

5.0 Internal Separating Elements

5.1 Internal Partition Walls

Table 5 below summarises the required rating of partition walls between teaching and other acoustically critical spaces.

Location of partition wall	Required acoustic rating (excluding glazing / doors)	Appraisal of proposed construction
Walls between classrooms / noise sensitive spaces and other rooms (no doors through)	Min. R_w 50dB	12.5mm Fermacell board with 10mm plywood backing. 75mm metal frame at 600mm centres with 60mm Rockwool flexi or equal insulation in the cavity. 12.5mm Fermacell board to other side.
Corridor walls containing doors	Min. R_w 45dB	12.5mm Fermacell board on 75mm metal frame at 600mm centres with 60mm Rockwool flexi or equal insulation in the cavity. 12.5mm Fermacell board to other side.
Walls separating staff rooms (offices, meeting room etc.)	Min. R_w 45dB	12.5mm wallboard (with plywood backing each side if required) on 50mm metal frame at 600mm centres with 25mm Rockwool flexi or equal insulation in the cavity.
Wall between plant room	Min. R_w 55dB	The existing wall between the plant room and motor workshop consists of 2 leaves 100mm solid blockwork with a 50mm cavity. As such, this should meet the required performance.

Table 4: Recommended partition wall specifications

5.2 Internal Doors and Glazing Specifications

Internal doors to noise sensitive rooms should have a minimum sound insulation value R_w 30dB. This would be achievable using standard FD30 solid core doors with the addition of acoustic perimeter and drop threshold seals. Glazed vision panels should not compromise the overall sound insulation of internal doors.

Based on the proposed layout at time of writing, access doors to the following rooms would require an acoustic rating min. R_w 30dB:

Rooms 2, 3, 4, 5, 6, 7, 10, 13, 14, 15, 20, 23, 25, 26, 28, 29, 30, 32, 33, 38, 41, 42, 43, 44, 46, 57, 62, 63, 64, 68, 69, 77, 81, 82, 84, 86, 87 & 91.

Group rooms / storerooms are generally considered to be non-teaching spaces and as such a min. rating of R_w 30dB should be sufficient for the interconnecting rooms.

BB93 states: *Where it is essential to link a teaching space with another occupied room via an interconnecting door for operational or safety purposes, a door set should be used with a rating of at least 35 dB R_w .* Group rooms are generally considered to be non-teaching spaces and as such a min. rating of R_w 30dB should be sufficient for the 3no. interconnecting rooms.

Internal glazing to the room numbers listed above should have a minimum sound insulation value R_w 33dB. This would suggest min. 8mm or 10mm laminated single glazing.

6.0 Plant and Ventilation Noise

6.1 Ventilation Noise

The following table provides a summary of internal noise limits for various rooms applicable to the scheme.

Room Type (Refurbishment)	Ventilation Type	Internal Noise Level Limit $L_{Aeq,30mins}$ dB
Assembly hall, multi-purpose hall	Natural	45dB
Secondary school: classroom, general teaching area, seminar room, tutorial room, language laboratory	Hybrid	Mechanical system only 40dBA
		Total level 45dBA
Office, medical room, staff room, meeting room etc.	Mechanical	45dB

Table 5: Indoor ventilation noise limits

It is understood that classrooms will be ventilated by means of a hybrid system incorporating Monodraught HTM FS units installed above windows. The following table provides an estimate of internal noise based on manufacturer quoted sound power data for 180l/s operation.

Item / Parameter	Octave Band Centre Frequency (Hz)								Total dBA
	63	125	250	500	1000	2000	4000	8000	
Monodraught HTM FS Sound power level @180l/s duty	52	55	53	47	40	32	22	23	49
Monodraught HTM FS Sound pressure level @1m distance	46	49	47	41	34	26	16	17	43
Predicted reverberant level, 180l/s operation, middle of typical classroom, RT approx. 1.1s	44	47	45	39	32	24	14	15	41
Predicted reverberant level, 180l/s operation, open window, middle of typical classroom, RT approx. 1.1s	46	47	45	40	35	30	20	15	42

Table 6: Ventilation noise estimate, single HTM FS unit serving typical classroom

Opening a window in teaching spaces would result in internal reverberant levels up to approx. 40dBL_{Aeq} (railway facing elevations), with combined levels from natural and proposed Monodraught system approx. 42dBA (assuming a single HTM FS unit; nominally add 3dB for an additional unit). As such, the levels would be compliant with targets set in BB93.

The new multi-purpose hall will be naturally ventilated by means of passive stack (wind catcher) ventilators at roof level. Based on the noise survey, ambient levels in the hall with wind catchers open would be expected to be <45dBA and as such this mode of ventilation would be seen as appropriate.

Offices, meeting rooms and other staff rooms will be ventilated by means of a mechanical (heat recovery) system. The mechanical heat recovery system should be designed to comply with the relevant parts of BB93; further review of the proposed system may be carried out when more detailed information is available.

6.2 Plant Noise

It is understood that the kitchen extract system is to be ducted to roof level, well away from any classroom or other noise sensitive room window. A noise limit of 70dBA @1m distance would be recommended for the roof level atmosphere connection to kitchen extract. The building is at significant distance from the nearest off-site receptor and as such it would be expected that a typical silenced kitchen extract system would comfortably comply with environmental noise limits for daytime periods. The proposed system may be reviewed when manufacturer noise data is identified.

Contents of the plant room will be reviewed when more detailed information is available.

7.0 Reverberation Control

This section provides a review of proposed surface finishes and recommendations on controlling reverberation within teaching and learning spaces in accordance with BB93 guidance. The objective is to provide suitable reverberation times (RT s) to facilitate:

- Clear communication of speech between teacher and student
- Clear communication between students
- Music teaching and performance

7.1 Multi-Purpose Hall

The proposed new multi-purpose hall is understood to be approx. 230m² area and 2000m³ total volume.

Proposed surface finishes in the new space can be summarised as follows:

- Cushioned vinyl floor
- Fair-faced blockwork walls to approx. 2.9m height
- Kingspan wall cladding (reverse of façade cladding panels) above 2.9m to soffit
- Profiled metal roof deck ceiling (understood lightweight insulated panel)

For multi-purpose halls (drama, PE, audio / visual presentation, assembly, occasional music), the target mid frequency (T_{mf}) reverberation time is defined as ≤ 1.5 seconds (refurbishment).

For sports halls, a reverberation time up to 2.0 seconds would be acceptable for refurbishment.

For new build schemes, the maximum Reverberation Time for sports halls reduces from 2 seconds for halls of floor area greater than 530 m² to 1.5 seconds for sports halls of floor area less than 280m².

Assuming the hall is used mostly for sports with only occasional presentations etc. an unfurnished target mid frequency RT range 1.5-2.0 seconds T_{mf} may be sufficient for the new hall.

Table 8 below details the proposed and recommended minimum acoustic surface finishes within the hall in order to comply with target reverberation time criterion.

Item / Parameter	Absorption coefficient approx. (Octave band, Hz)						Average T_{mf} 500-2kHz
	125Hz	250Hz	500Hz	1kHz	2kHz	4kHz	
Walls lower: Fair faced blockwork	0.05	0.05	0.05	0.10	0.15	0.20	
Walls upper: Lightweight insulated panel metal finish	0.20	0.15	0.10	0.05	0.05	0.05	
Floor: Cushioned vinyl	0.05	0.05	0.10	0.10	0.05	0.05	
Ceiling: Trapezoidal profiled lightweight panel metal finish	0.20	0.15	0.10	0.05	0.05	0.05	
Recommend additional absorption: 15% of wall & ceiling area (nom. 120m ²) acoustic panels (min. Class C absorber type)	Recommended minimum α (Class C or better)						
	0.2	0.5	0.8	0.8	0.8	0.7	
Predicted Reverberation Time RT (seconds)	2.0	2.0	1.8	1.9	1.8	1.4	≈1.8s

Table 7: Schedule of recommended acoustic treatments, new multi-use hall

As such, approx. 120m² of Class C absorber type wall panel should be deemed sufficient in order to comply with BB93 targets for reverberation in refurbished multi-purpose halls of this size. Examples of suitable products (impact resistant) would include:

- Soundsorba Steelsorba
- Soundsorba Toughsorba

Panels should be installed with the minimum cavity behind to achieve absorption performance, as per manufacturer guidelines.

Panels should be positioned in upper areas of walls (see Figure 2 below indicated in green) and generally across the ceiling.

Alternatively, perforation of the Kalzip roof underliner could be considered, which would provide additional acoustic absorption, thus eliminating the need for ceiling panels and reduce the required wall panel area. This option could be assessed in more detail, given perforation area data from Kalzip.

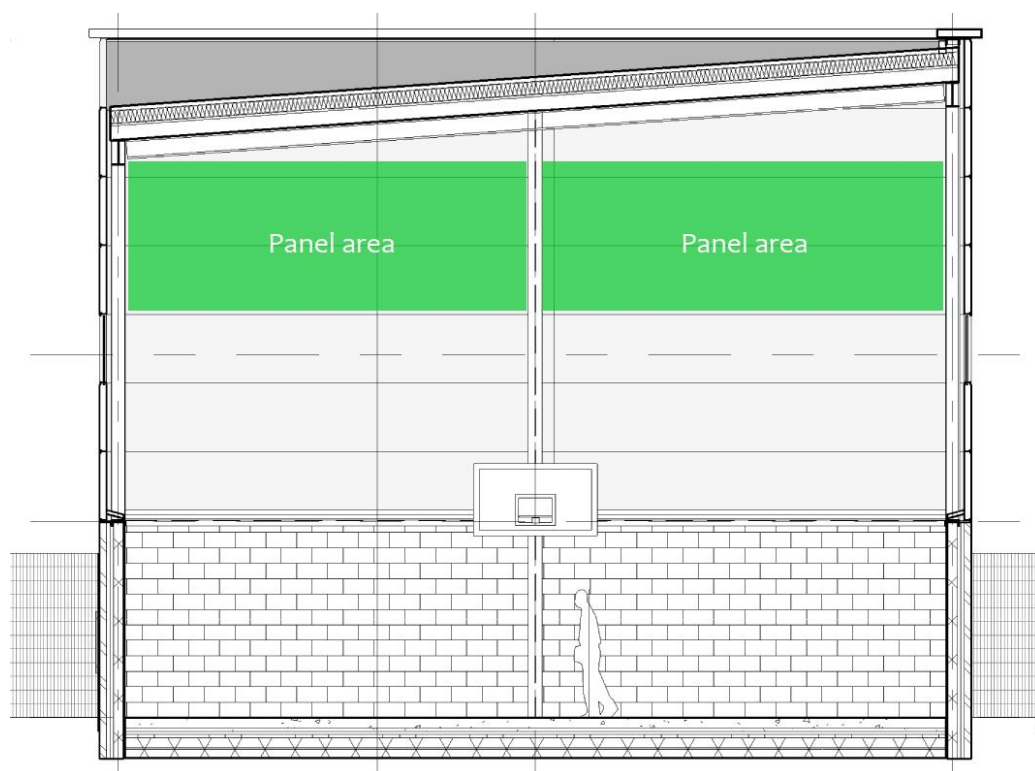


Figure 2: Section through hall indicating recommended location of acoustic wall panels (green)

7.2 General Teaching Accommodation

This section outlines the required surface finishes required for specific spaces within the building, with the aim of achieving suitable target reverberation times as specified in BB93 guidance.

The following table assumes a nominal ceiling height of 3.85m in all rooms and thus would need to be reviewed if ceiling heights in some areas are to be lower.

Floor finishes are assumed, based on experience of similar projects (no information provided) and as such would be subject to review if plans differ.



Room	Volume ≈m ³	Target Reverb Time RT (≤ sec)	Assumed Floor	Proposed Ceiling	Additional treatment recommended
46 - KS2 Classroom	153	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
43 - KS2 Classroom	163	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
87 - KS3 Classroom	184	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
86 - KS3 Classroom	193	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
81 - KS3 Art Classroom	215	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
29 - KS2 Classroom	177	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
26 - KS2 Classroom	172	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
30 - KS2 Classroom	174	1.0	Carpet	Plasterboard	10m ² acoustic wall / ceiling panels
64 - Science Lab	204	1.0	Vinyl / hard	Plasterboard	26m ² acoustic wall / ceiling panels
70 -Dining Area	371	1.5	Vinyl / hard	Plasterboard	16-18m ² acoustic wall / ceiling panels
20 – Library Room	107	1.2	Carpet	Plasterboard	2-4m ² acoustic wall / ceiling panels
6 - Conference Meeting	77	0.8	Carpet	Plasterboard	6m ² acoustic wall / ceiling panels
7 - Head Office	54	1.2	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
3 - Office Manager	30	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet
91 - Meeting Room	38	0.8	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
8 - Prem Manager	19	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet
5 - Staff Wellbeing	30	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet
4 -Safeguarding	27	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet
2 - Gen. off Room	77	1.2	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
44 - Group Room	77	1.0	Carpet	Plasterboard	6m ² acoustic wall / ceiling panels
82 - Cube Room(s)	30	0.8	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
77 - Office Room	42	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet
41 - Counselling	38	0.8	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
42 - Sensory Room	46	0.8	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
37 - Staff Room	136	1.2	Carpet	Plasterboard	4m ² acoustic wall / ceiling panels
34 - Staff Work	46	1.2	Carpet	Plasterboard	Generally sufficient assuming carpet



23 - Food Room	226	1.0	Vinyl / hard	Plasterboard	Carpet tiles or 28m ² acoustic wall / ceiling panels
Circulation	92	1.5	Vinyl / hard	Ceiling tiles	Class C absorber type mineral tile
Circulation	287	1.5	Vinyl / hard	Ceiling tiles	Class C absorber type mineral tile
Key				Blue	Recommended for good acoustics
				Red	Required for compliance with BB93: 2015

Table 8: Schedule of recommended acoustic treatments, new / refurbished accommodation

The proposed acoustic wall / ceiling panels are understood to be Ecophon Super G Class A panels, and as such should be deemed sufficient.

Incorporation of the surface finishes described in the table above would result in compliance with Table 6 of Building Bulletin 93 and would generally provide a good acoustic environment.



Appendix A – Glossary of Terms

Decibel (dB)

this is the unit used to measure sound. The human ear has an approximately logarithmic response to acoustic pressure over a very large dynamic range (typically 20 micro Pascal to 100 Pascal).

dB (A)

This is a measure of the overall noise level of sound across the audible spectrum with a frequency weighting (i.e. A-weighting) to compensate for the sensitivity of the human ear to sound of different frequencies. The A-weighting curve is implemented in sound level meters using an electronic filter that approximately corresponds to the frequency response of the ear.

Octave Band Noise Level

The human ear is sensitive to sound over a range of frequencies between approximately 20 Hz to 20 kHz. The ear is also generally more sensitive to medium and high frequencies than to low frequencies. In order to define the frequency content of a noise, the spectrum can be divided into frequency bands. The most commonly used frequency bands are octave bands, in which the mid-frequency of each band is twice that of the band below it.

L_{Aeq}

this is the equivalent steady sound level in dB (A) containing the same acoustic energy as the actual fluctuating sound level over a given time period.

Reverberation Time (RT or sometimes T_{30} or T_{60})

This is the time taken for the reverberant sound energy in an enclosure to decay one millionth of its equilibrium value, i.e. by 60 dB, after the source has been switched off, is known as the reverberation time. The reverberation time is frequency dependent and it is customary to measure its value in octave or one-third octave bands. Reverberation occurs when sound waves are repeatedly reflected from each surface of the room.

Sound Reduction Index (SRI)

Difference measured between the amount of energy flowing towards the wall in the source room and the total amount of energy flowing towards the wall in the source room and the total amount of energy entering the receiving room (usual range 100 - 3150 Hz for one third octave band values). The SRI varies with frequency and is measured in a laboratory in either octave or one-third octave bands.

$$SRI = L1 - L2 + 10 \log (S/A)$$

Where: $L1$ = Noise level in the source room

$L2$ = Noise levels in the receiving room

S = Surface area of test specimen

A = Equivalent acoustic absorption area in the receiving room

Weighted Sound Reduction Index (R_w)

this is a weighted single figure descriptor of the sound insulation performance of a partition measured under laboratory conditions. The procedure used to quantify the R_w is to compare the sound reduction index (SRI) in each of the one-third octave bands from 100Hz to 3150Hz against a set of standard reference curves.

Appendix B – Graphical Results of Sound Insulation Tests